

Mingli Zhang

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Objective

Machine Learning Backend Engineer specializing in LLM fine-tuning, RAG pipelines, computer vision, and AI agent deployment — focused on output quality and real-world impact. Backed by the backend infrastructure that keeps every service reliable and consistently accessible at scale.

Work Experience

ByteDance | Machine Learning Backend Engineer | San Jose, CA, USA

Jan 2024 - Present

IT Enterprise Solution AI Operations Algorithm Team

- Fine-tuned LLMs with **LLaMA-Factory** on self-generated network fault data using **SFT and LoRA**, improving diagnostic report quality and accelerating RCA workflows for operations teams.
- Architected a real-time server room surveillance system using computer vision, deployed across **140+ sites** and expanding to **1,200+**, that automatically detects unauthorized human presence and behavior — eliminating the need for manual monitoring.
- Built a three-stage CV pipeline: **YOLO**-based body detection (**<1s**), **YOLO-Pose** with positional calculations for intelligent key frame extraction, and a self-deployed **Qwen3-VL-MoE** vision model to classify unauthorized behaviors (e.g. eating, drinking) — analyzing 10-second clips in under **5 seconds**.
- Built a custom **FastAPI**-based inference server with multi-threaded HTTP workers and batched GPU workers, doubling QPS capacity and **improving GPU utilization by 60%**, with middleware for event logging and distributed tracing integrated into ByteDance's observability systems.
- Built a Text-to-SQL service with **LangChain** and **RAG** that enables operators to query network data in plain language, saving **100+ human hours monthly** across all offices.
- Patented an algorithmic approach and co-authored 3 additional patents on machine learning solutions for real-world infrastructure problems.

Cymantix | Software Engineer (Part Time) | North Carolina (hybrid), USA

Oct 2021 - Feb 2023

A tech startup integrating machine learning with data analytics and information retrieval.

- Collaborated with 5 developers to build a **machine-learning-driven platform** integrating WebRTC video conferencing, information retrieval, and a core identity API — contributing to the product that secured seed-round funding.
- Improved clustering quality by replacing **K-Means** with **Hierarchical Clustering**, reducing user wait time by **10%–60%** depending on dataset size.
- Optimized real-time news analysis using **multithreading**, **Node.js**, and **Socket.io**, cutting user wait time by **90%**.
- Integrated **Microsoft Azure** for authentication and cloud storage, enabling a cost-effective platform transition that brought in **2 non-profit clients within 1 week**.

IBM | Full-Stack Engineer (Summer Intern) | North Carolina, USA

May 2021 - Aug 2021

Worked on IBM's internal platform that provisions and customizes live demo environments.

- Enhanced the experience of **12,000+ IBMers monthly** using **Next.js** by implementing intuitive improvements driven directly by user feedback.
- Built an automated validation service using **Sails.js** that streamlined database maintenance and saved the team **\$14,000 annually** by eliminating manual tasks.
- Streamlined administrative content workflows, enabling faster updates and reducing developer redeployments by **20+ per week**.

Education

University of North Carolina at Chapel Hill

Master's of Science in Computer Science | Aug 2022 - Aug 2023

- **GPA: 3.9/4.0.**
- **Courses:** Advanced Machine Learning, Digital Logic, Advanced Operating System.
- **Research:** Key-Value Store with researchers from VMWare using **x86** and **C**.

Bachelor's of Science in Computer Science and Mathematics | Aug 2018 - May 2022

- **GPA: 3.8/4.0**, Dean's List from 2018-2022.
- **Courses:** Algorithms and Analysis, Files and Database, Operating System, 2D Computer Graphics, Quantum Computing

Skills

Languages Python, Golang, C/C++, JavaScript

ML & Training PyTorch, LLaMA-Factory, Hugging Face, scikit-learn, LoRA, SFT, CUDA

LLM & CV LangChain, RAG, YOLO, YOLO-Pose, Agent-SDK

Infrastructure & Systems Docker, Kubernetes, FastAPI, gRPC, SQL, Linux, CronJob